

Brain Tumor Classification Using Deep Learning and 3D Transformer Integration

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Abstract—Brain tumors are one of the most serious neurologic diseases and their timely and correct diagnosis is essential for good treatment planning. Magnetic Resonance Imaging (MRI) is a widely used technique for brain tumor detection because of its ability to provide excellent soft-tissue contrast and is a non-invasive technique. In this study, a comprehensive brain tumor classification model based on MRI images in the BraTS dataset is presented. The proposed system incorporates image preprocessing, tumor segmentation, feature extraction, traditional machine learning, deep learning, and Transformer-based learning. Gaussian, mean, median filtering, anti-aliasing and resizing methods were used to enhance the images obtained from MRI. In tumor regions, the segmentation was performed by Sobel edge detection and morphological operations. Texture and geometric features were extracted and used for classification. Three classification methods were tested: Random Forest, Convolutional Neural Network (CNN) and a 3D CNN-Transformer architecture. The experimental results demonstrated that the accuracy of the Random Forest classifier was 87.5%, while the CNN and the Transformer-based model achieved 80% and 40% accuracy, respectively. The results show that the traditional machine learning and CNN-based methods can perform well when the dataset is small, while the Transformer architectures generally need a larger dataset to effectively take advantage of their attention mechanisms.

I. INTRODUCTION

Brain tumors are uncontrolled growths of cells in the brain which can have important neurological and life-threatening implications. Their growth characteristics and aggressiveness can be used to determine if they are benign or malignant. The early and accurate diagnosis is crucial for effective treatment planning and better patient outcomes.

One of the most common imaging methods used for the diagnosis of brain tumors is Magnetic Resonance Imaging (MRI) which avoids the use of ionizing radiation and provides detailed anatomic information. Yet, manual review of MRI images is time-consuming and can be very reliant on radiologist expertise.

With the advent of recent Artificial Intelligence (AI) and Deep Learning (DL) breakthroughs, automated systems have been developed to help physicians with disease diagnosis and medical image interpretation. Deep learning models can learn the discriminative image features and recognize complex patterns which are not easily captured by conventional image analysis.

The classification of medical images is a prominent area of image processing that has been successfully tackled by Convolutional Neural Networks (CNNs) by directly learning the hierarchical spatial features of

images. Recently, Transformer architectures have proven to be strong alternatives that are able to capture long-range dependencies by leveraging self-attention mechanisms. Transformers are very good in representing global context; they learn the relationship between all image points or image volumes, while CNNs are effective in extracting local image features.

The purpose of this study is to design a whole brain tumor classification system, including image preprocessing, segmentation, feature extraction, CNN classification, and 3D Transformer fusion. The effectiveness and comparison of Random Forest, CNN-based methods and Transformer methods are assessed with MRI images from BraTS dataset.

II. LITERATURE REVIEW

MRI is now the preferred imaging modality for diagnosis of brain tumors due to its excellent soft tissue contrast and ability to differentiate tissue structures. Researchers have been able to create and test automated brain tumor detection and classification systems using public datasets like the Brain Tumor Segmentation (BraTS) dataset. The techniques used for traditional image processing are explored.

In medical image analysis, the traditional image processing techniques are well known. Simple methods include thresholding, edge detection, morphology and texture analysis. All these methods are computation-efficient and are applicable to deriving meaningful structural information from medical images. Their performance is, however, limited in cases where the tumor borders are irregular and/or the intensity of the image's changes.

2.3 Deep Learning for Medical Imaging

Medical image analysis has been revolutionized by deep learning, which can automatically learn features. Convolutional Neural Networks (CNNs) show very good performance in particular because they are able to learn hierarchical representations of images directly from the raw image data.

Deepak and Ameer (2019) have proposed the deep CNN transfer learning based brain tumour classification framework. Their study showed that CNNs can automatically learn discriminative MRI features and produce high classification accuracy even when no much manual feature engineering is needed.

The key benefits of CNNs are:

- Automatic feature extraction
- Less reliance on manually created features
- Strong classification performance
- End-to-end learning capability

2.4 Transformer-Based Architectures

Inspired by Natural Language Processing (NLP), the transformers were originally developed for NLP tasks and have recently proven to be very successful in computer vision. Transformers, unlike CNNs, rely on self-attention mechanisms to capture relationships between remote areas in the image.

Asiri et al. (2023) explored Vision Transformers (ViT) for brain tumor classification and showed that the self-attention mechanisms could learn the global-contextual relationships between MRI volume components.

Transformers come with a number of benefits, such as:

- Global feature representation
- Long-range dependency modeling
- Improved contextual understanding
- Flexible feature learning

2.5 Hybrid CNN-Transformer Models

Recently, hybrid architectures employing CNNs and Transformers have been investigated. CNN is used for local spatial feature extraction, and Transformer blocks are used to model global relations through self-attention.

Aloraini et al. (2023) suggested a CNN-Transformer based brain tumor classification model using MRI images. They found that self-attention combined with convolutional operations can enhance the feature representation and classification results.

The above studies have shown that CNNs can be useful for local features extraction, and Transformers can be used for modeling global features context. Hence, the combination of both methods could be beneficial for medical image classification.

III. METHODOLOGY

3.1 Dataset

The BraTS dataset was used in this study. This data set consists of multimodal MRI scans and corresponding tumor segmentation masks.

These are the different modalities offered:

- T1-weighted MRI
- T1 Contrast Enhanced MRI (T1c)
- T2-weighted MRI
- FLAIR MRI
- Tumor Segmentation Masks

Twenty-five patient cases were picked and subjected to experimentation.

3.2 Image Preprocessing

Image preprocessing was done to enhance the image quality and noise removal prior to segmentation and classification.

Gaussian Filtering

Gaussian filtering was used to smooth the variations of intensity in the image and reduce the random noise.

Mean Filtering

Mean filtering helped to further reduce noise in images by taking the average of the neighboring pixel intensity.

Median Filtering

Impulse noise was eliminated and important edge information was preserved by median filtering.

Anti-Aliasing and Resizing

To prevent image distortion and to maintain the quality of the image, anti-aliasing was used prior to downsampling.

3.3 Tumor Segmentation

Edge-based image processing techniques were used for tumor segmentation.

The segmentation pipeline consisted of the following:

- Sobel Edge Detection
- Binary Thresholding
- Erosion
- Dilation
- Opening
- Closing

Small artifacts were removed from segmentation masks and the tumor boundary was better extracted using morphological operations.

3.4 Feature Extraction

On segmented regions, texture and geometrical characteristics were extracted from tumor regions.

Texture Features (GLCM)

The following features were computed by using the Gray Level Co-occurrence Matrix (GLCM):

- Energy
- Entropy
- Contrast
- Geometric Features

Geometric properties of the tumor were calculated such as:

- Area
- Perimeter
- Circularity
- Centroid Coordinates

These features were recorded in a csv file and applied to a traditional machine learning classification.

3.5 Random Forest Classification

Texture and geometric features were extracted and a Random Forest classifier was trained with these features. Tumors were divided into two categories: High Risk, and Low Risk, according to the median tumor area.

Random Forest was chosen due to its robustness against overfitting and small datasets.

3.6 CNN Classification

A Convolutional Neural Network (CNN) was designed to classify tumour images.

The following changes were made to boost performance and decrease overfitting:

- Data Augmentation
- Batch Normalization
- Early Stopping
- Dropout Regularization
- Larger Convolution Filters (5×5)

The CNN architecture was composed of:

1. Data Augmentation Layer
2. Conv2D (32 Filters, 5×5)
3. Batch Normalization
4. Max Pooling
5. Conv2D (64 Filters, 5×5)
6. Batch Normalization
7. Max Pooling
8. Conv2D (128 Filters, 5×5)
9. Batch Normalization
10. Max Pooling
11. Dense Layer (128 Neurons)
12. Dropout Layer (0.5)
13. Sigmoid Output Layer

Image rotation, zooming, flipping and translation were applied to the data.

3.7 3D CNN-Transformer Architecture

A 3D CNN architecture was used to integrate a Transformer block to include advanced deep learning techniques.

The complete MRI volume was used instead of individual image slices. Tumor mask was created to isolate tumor areas which were then resized to 64×64×64.

The architecture included:

- 3D Convolution Layers
- 3D Max Pooling Layers
- Multi-Head Self-Attention
- Layer Normalization
- Dense Layers

- Dropout
- Sigmoid Output Layer

The Transformer architecture was used to model voxel-to-voxel dependencies, and to capture global spatial relationships between voxels in the MRI volume, with the use of a Multi-Head Self-Attention mechanism.

To make the training more stable and less prone to overfitting, Batch Normalization and Early Stopping were added.

3.8 Performance Evaluation

We tested the proposed framework on three classification methods: Random Forest, Convolutional Neural Network (CNN) and a 3D CNN-Transformer architecture.

The extracted texture features from GLCM and geometric features were taken as input for Random Forest classification. The tumors were divided into two groups: High Risk (1 group) and Low Risk (2 group), according to the median tumor area. The main advantage of the proposed handcrafted texture and shape features and their effectiveness in classification of brain tumor is shown from the accuracy obtained by the Random Forest (87.5%).

Processed MRI images were used to train the CNN model. To enhance the generalization and prevent overfitting, data augmentation, batch normalization, dropout, and early stopping were added. The CNN performed at an accuracy of 80.0% on the validation dataset - having correctly classified 4 of the 5 samples in the validation set.

A 3D CNN-Transformer architecture that utilizes multi-head self-attention was designed to model global voxel relationships in MRI volumes. The model outperformed the others, with an accuracy rate of **40.0%**, even though it is a transformer-based model. The smaller size of the data set may be the reason for the poorer performance; typically, transformer architectures require a much larger data set to be properly trained.

The overall classification performance of all the models evaluated are summarized in Table 1.

Table 1. Classification Performance Comparison

Model	Accuracy
Random Forest	87.5%
CNN	80.0%
3D CNN + Transformer	40.0%

It can be seen from the results that the Random Forest classifier is the highest performing model on the selected dataset followed by CNN model. While the Transformer architecture is highly capable of learning global dependencies by means of powerful attention mechanisms, the usage was constrained due to the limited number of MRI samples available.

IV. CONCLUSION

This research proposed an entire brain tumor classification model for MRI images of the BraTS dataset. The framework incorporated Preprocessing, Segmentation, Feature Extraction, Random Forest classification, CNN classification and Transformer-based learning.

To enhance the quality of MRI, image preprocessing was performed by applying filtering and resizing operations and to perform tumor segmentation Sobel edge detection and morphological operations were applied. The texture and geometric features were extracted and employed in the traditional machine learning classification.

Three classification methods were tested: Random Forest, CNN and 3D CNN-Transformer. The highest accuracy of 87.5% was seen in the Random Forest and CNN with an accuracy of 80.0%. With the small data size, the accuracy of the Transformer-based model was 40.0%.

The results show that traditional machine learning and CNN-based methods are still very good in small medical imaging datasets. Though Transformers have strong feature modeling capabilities globally, they are very reliant on having large training sets. In future, larger MRI datasets, transfer learning, and more sophisticated Vision Transformer architectures can be explored to further enhance the classification results.

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